

3.1 Troubleshooting Display Problems

Earlier, images displayed on the monitor used to be processed by the CPU and sent to the monitor via the video adapter (also known by various other names such as display adapter, video card, video board, video display board, graphics card, graphics adapter etc). To change the image on the screen, a computer needs to change individual pixels. GUIs (Graphical User Interfaces) and image-



A motherboard with on-board graphics

intensive applications increased the burden on the CPU and the RAM, causing a performance hit on the rest of the system. To overcome this bottleneck, accelerated graphics cards were introduced. These took over the image processing tasks from the CPU, and, as is common in technology, graphics cards have become complex sub-systems in themselves. A typical mid-range graphics card is like a mini-motherboard with its own Graphics Processor, memory (RAM) and even a heat sink for all the heat generated. With growth in complexity came the increase in price, which led to a kind of reverse innovation where motherboard manufacturers began to incorporate the graphics card directly onto the motherboard. These are generally known as 'on-board' or integrated graphics adapters, and give fairly good graphics handling capability for general use.

Troubleshooting the video subsystem requires a basic understanding of what kind of graphics card you have—on-board or